Advancing Food Nutrition Estimation via Visual-Ingredient Feature Fusion

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ABSTRACT

Accurate estimation of food nutrition from images is a critical step toward automated dietary monitoring and public health management. However, existing approaches are bottlenecked by two key challenges: the scarcity of large-scale datasets with paired ingredient and nutrition annotations, and an over-reliance on portion-size or depth cues that do not generalize well across diverse food categories. In this paper, we address both challenges with two complementary contributions. First, we introduce **FastFood**, a large-scale benchmark of 84,446 images spanning 908 fast-food categories from 10 global brands, each annotated with precise calorie, fat, carbohydrate, and protein values, as well as detailed ingredient lists. Second, we propose **VIF**^{2} (**V**isual-**I**ngredient **F**eature **F**usion), a model-agnostic method that encodes ingredient semantics via CLIP's text encoder and injects the resulting embeddings into intermediate visual representations of CNN or ViT backbones, guiding multi-task nutrition regression. At test time, ingredients are predicted by large multimodal models (LMMs) from augmented views with majority voting to ensure robustness. Experiments on FastFood and Nutrition5k demonstrate substantial gains: on FastFood, VIF^2 reduces ResNet101's caloric MAE from 118.04 to 61.26 (a 48.1% reduction), with consistent improvements across all macronutrients and backbone architectures.

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1 INTRODUCTION

Food nutrition estimation from images is a foundational capability for automated dietary assessment systems [20], holding significant implications for personal health management [3], clinical nutrition monitoring [17], and sustainable dietary planning [28]. Despite the importance of the problem, it remains fundamentally difficult: food appearance varies widely across preparation styles and portion sizes, and most datasets that support nutrition estimation are either small in scale, limited in category diversity, or lack ingredient-level annotations.

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Early work on food recognition focused on classification over fixed category sets [2, 5], while calorie estimation methods typically relied on portion size estimation via depth sensors or 3D reconstruction [10, 27, 29]. The Nutrition5k dataset [27] is the most widely used benchmark for nutrition regression, but it covers only generic home-cooked dishes with limited scale, and its nutrient annotations are obtained by weighing individual ingredients, making collection expensive. Recent approaches have used depth maps [10, 25] or multi-task learning [24] to improve estimation, but these strategies rely on equipment unavailable in casual settings or on recipe data without fine-grained nutrient labels.

A key observation motivating this work is that *ingredients are the primary determinant of nutritional content*: knowing that a dish contains cheese, beef patty, and a sesame bun immediately constrains its macronutrient profile far more tightly than visual appearance alone. Ingredient prediction from food images has been studied independently [1, 18, 21], and ingredient information has been used for recipe retrieval [4] and recipe generation [6], but its explicit use to guide nutrition regression has remained largely unexplored.

We address this gap with two contributions. First, we construct **FastFood**, a large-scale fast-food nutrition dataset with 84,446 images across 908 categories from 10 global brands, annotated with calories, fat, carbohydrates, protein, and standardized ingredient lists. To our knowledge, FastFood is the largest nutrition-labeled food image dataset with ingredient annotations. Second, we propose **VIF**^{2} (**V**isual-**I**ngredient **F**eature **F**usion), which projects CLIP-encoded ingredient text embeddings into the visual feature space and fuses them into intermediate layers of CNN or ViT backbones for multi-task nutrition regression. At inference time, when ground-truth ingredients are unavailable, we use large multimodal models (LMMs) to predict ingredients from multiple augmented views with majority voting, enabling robust ingredient-guided prediction.

Our experiments demonstrate that VIF^2 consistently and substantially improves all backbone architectures on both FastFood and Nutrition5k. On FastFood, ResNet101 with VIF^2 achieves a caloric MAE of 61.26 compared to 118.04 for the baseline (48.1% reduction). On Nutrition5k Protocol 2, ResNet101 with VIF^2 achieves 23.32 caloric MAE, a strong result enabled by the additional ingredient signal. Ablation studies confirm that intermediate-layer fusion and the majority-voting ingredient strategy are key design decisions. The main contributions of this paper are:

- **FastFood dataset**: a large-scale fast-food benchmark with 84,446 images across 908 categories, annotated with calories, fat, carbohydrates, protein, and standardized ingredients.
- **VIF**^{2}: a model-agnostic visual-ingredient feature fusion module that injects CLIP-encoded ingredient semantics into CNN or ViT backbones for multi-task nutrition regression.

- **LMM-driven ingredient inference**: an augmented view prediction strategy with majority voting for test-time ingredient estimation without ground-truth labels.
- Comprehensive evaluation on FastFood and Nutrition5k, including ablations of fusion layers, LMM choices, and ingredient robustness strategies.

2 RELATED WORK

2.1 Food Recognition and Calorie Estimation

Food recognition has progressed rapidly from shallow feature methods to deep learning approaches [2, 5, 14, 16]. Large-scale food datasets such as Food-101 [2] enabled training high-accuracy classifiers, but these datasets lack nutrition labels. Calorie estimation approaches typically combine food recognition with portion size estimation. Early methods used RGB-D cameras [27] or stereo vision to estimate food volume, then looked up calorie density. More recent deep learning approaches train end-to-end regression models [9, 19], sometimes using multi-task learning to predict multiple nutrients simultaneously [24]. The Nutrition5k dataset [27] enables nutrition estimation from overhead RGB and depth images. Subsequent work has proposed depth prediction modules [10, 29], RGB-D fusion networks [25], and transformer-based approaches [30] to improve estimation on Nutrition5k. Our work differs by exploiting ingredient semantics as a complementary signal to visual features, achieving substantial gains without any depth information.

2.2 Ingredient Recognition

Recognizing ingredients from food images is a well-studied multi-label classification problem [1, 18, 21]. Ingredients have been used to bridge food images and recipes for retrieval [4] and generation [6, 8]. Recent work has used graph neural networks [21] and attention mechanisms [18] to improve ingredient recognition. VIF^2 differs from these works in that we use ingredients not as a prediction target but as a conditioning signal for nutrition regression.

2.3 Multi-Task Learning for Food Understanding

Multi-task learning has been widely adopted in food computing to jointly predict multiple nutritional attributes [24]. Sharing intermediate representations across calorie, fat, carbohydrate, and protein estimation encourages the model to learn generalizable visual features rather than overfitting to any single nutrient. VIF^2 builds on this paradigm by introducing ingredient semantics as a shared conditioning signal across all regression heads, infusing compositional knowledge that visual features alone cannot capture. In practice, calorie values (hundreds of kilocalories) are an order of magnitude larger than macronutrient masses in grams, causing naive joint training to prioritize the caloric loss at the expense of macronutrient accuracy. VIF^2 addresses this with explicit task weighting (λ_{cal}=0.01, λ_n=1 otherwise) while using the CLIP-encoded ingredient embedding as a shared nutritional prior that simultaneously guides and regularizes all four regression heads toward compositionally coherent predictions.

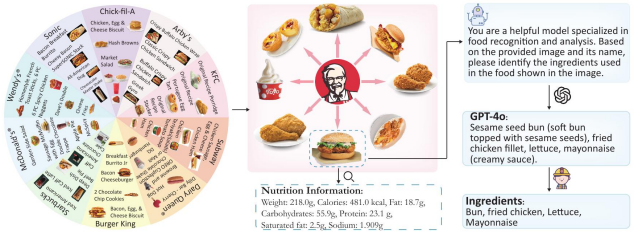

Figure 1: Overview of the FastFood dataset, showcasing brands, sample images, and annotated ingredients and nutrition values.

2.4 Cross-Modal Feature Fusion

Cross-modal feature fusion has been widely studied in visual question answering, visual grounding, and image captioning, where conditioning visual representations on language context improves task performance [23]. FiLM-style mechanisms [22] inject language features into visual pipelines via learned affine transformations, while cross-attention allows visual tokens to selectively attend to relevant textual context. In the food domain, cross-modal approaches have primarily targeted recipe-image retrieval [4], aligning visual and text encoders in a shared embedding space. VIF^2 draws on this tradition but pursues a different objective: rather than aligning modalities for retrieval, we inject ingredient semantics as a conditioning signal into a discriminative visual regression backbone, directly exploiting the causal relationship between ingredient composition and macronutrient content.

2.5 Vision-Language Models for Food

Large vision-language models have been applied to food understanding in several recent works. FoodLMM [32] is a food-specific LMM supporting nutrition estimation, ingredient recognition, recipe generation, and segmentation. RoDE [12] proposes a mixture-of-experts approach for multi-task food understanding. CLIP [23] has been used for zero-shot food recognition and cross-modal retrieval. We leverage CLIP's text encoder to produce rich ingredient embeddings and inject them into discriminative visual backbones, combining the semantic richness of language models with the efficiency of task-specific CNN/ViT regression. LLaVA [15] and similar LMMs are used at test time for ingredient prediction.

3 FASTFOOD DATASET CONSTRUCTION

3.1 Overview

FastFood is a large-scale fast-food nutrition dataset designed to support nutrition estimation from food images. It contains **84,446 images** across **908 food categories** from **10 global fast-food brands**, each annotated with calories, fat, carbohydrates, protein (in grams), and a standardized ingredient list. Figure 1 shows an overview of the dataset and illustrative examples.

Compared to Nutrition5k [27], FastFood offers broader category coverage, standardized brand-level nutrition labels (rather than ingredient-weight-derived estimates), and an order-of-magnitude more images. The dataset is split 70/20/10 for training, validation, and testing.

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Figure 2: Data expansion for food items using text-based and image-based retrieval methods to increase diversity.

3.2 Data Collection and Expansion

Nutrition information and seed images are sourced from official brand websites, ensuring precise and authoritative nutrition labels. To increase image diversity beyond official product photos, we expand each category using two complementary retrieval strategies (Figure 2):

- **Text-based retrieval**: Google image search using food item names to capture diverse real-world photographs.
- **Image-based retrieval**: Reverse image search from seed images to retrieve visually similar clean product images.

3.3 Filtering and Quality Control

Collected images are filtered to remove: blurry or low-resolution images, images showing incorrect item sizes or multiple unrelated food items (except combo meals), and images with significant visual mismatch from the target item. Nutrition values are standardized to a consistent unit system (kcal, grams).

3.4 Ingredient Annotation

For each food item, we construct a standardized ingredient list using GPT-4o [13] as an initial proposer: the model is prompted with the item name and brand to generate a plausible ingredient list. Human annotators then refine the proposed lists by: (i) enforcing specificity (e.g., replacing "vegetables" with "lettuce, tomato, onion"); (ii) removing duplicate or irrelevant entries; (iii) standardizing ingredient names across the vocabulary.

Manual verification of both images and ingredient lists is performed before finalizing the dataset.

3.5 Dataset Statistics

Figures 3 and 4 show the distribution of calorie and macronutrient values. Calorie values span a wide range, reflecting the diversity of fast-food items from small snacks to large combo meals. Figure 5 shows the top-20 most frequent ingredients, dominated by common fast-food staples such as salt, flour, and vegetable oil.

Additionally, 5,800 images in FastFood are enriched with 20,204 multi-turn dietary planning dialogues generated by GPT-4o, making the dataset suitable for future research on diet recommendation and conversational food AI (Figure 6).

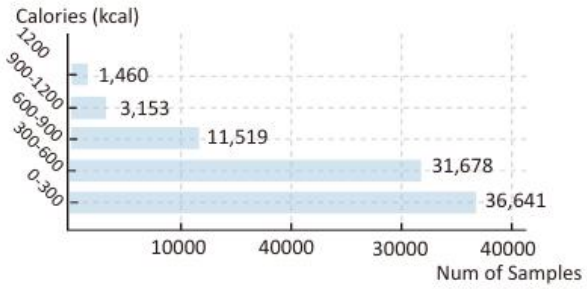

Figure 3: Distribution of calorie values in the FastFood dataset.

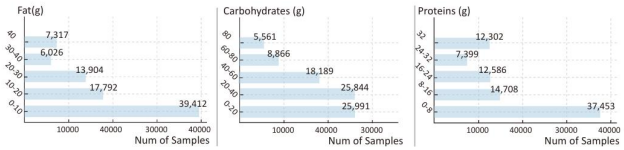

Figure 4: Distribution of fat, carbohydrate, and protein content across the FastFood dataset.

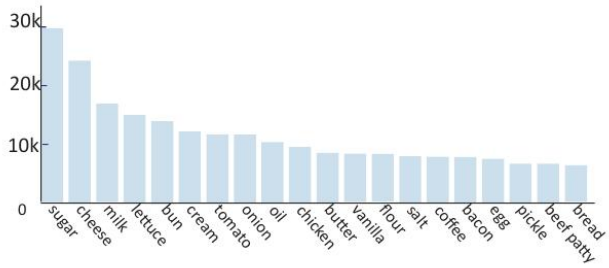

Figure 5: Top-20 most frequent ingredients in the FastFood dataset.

4 METHOD: VIF^2

4.1 Overview

VIF^2 (Visual-Ingredient Feature Fusion) is a model-agnostic module that injects ingredient semantics into the visual feature extraction pipeline of a backbone network for multi-task nutrition regression. The key idea is to encode the ingredient list of a food item using CLIP's text encoder [23], project the resulting embedding into the visual feature space, and fuse it into intermediate feature representations. This allows the backbone to attend to visually ambiguous regions differently depending on the known ingredients, producing more accurate nutrition estimates. Figure 7 illustrates the full pipeline.

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...I will provide a fast food image and its nutrion informaon. Please help me generate a multi-round dialogue about the image on dietary planning and suggestions as the model's training data.
For example, nutrition information description like this: TOTAL(194, 15.12, 0.5, 13.3) means this food item has 194 kcal calories and 15.12g fat, 0.5g carbohydrate and 13.3 protein.
One 1-round example is: \Image Nutrition Information: TOTAL (500, 23, 49, 24) **USER:** Can you provide some dietary suggestions based on this image? **ASSISTANT:** Sure! To balance this meal, consider pairing it with low-calorie, nutrient-rich side such as a green salad...
Nutrition Information: TOTAL ({cal}, {fat}, {carb}, {pro}). Please help me generate a {N}-round dialogue closely related to dietary planning and suggestions. The questions are expected to be professional or general, the answers are expected to be explanatory.

Figure 6: Prompt template used for generating multi-turn dietary planning dialogues.

4.2 Ingredient Encoding

Given a food item with ingredient list $\mathcal{I} = \{i_1, i_2, \ldots, i_K\}$, each ingredient name i_k is encoded using CLIP's frozen text encoder f_{CLIP}:

$$\mathbf{e}_k = f_{\text{CLIP}}(i_k) \in \mathbb{R}^d. \tag{1}$$

The per-ingredient embeddings are aggregated by mean pooling to obtain a single ingredient feature vector:

$$\mathbf{e}_{\mathcal{I}} = \frac{1}{K}\sum_{k=1}^{K} \mathbf{e}_k \in \mathbb{R}^d. \tag{2}$$

A lightweight projection head g_θ (linear layer followed by layer normalization) maps $\mathbf{e}_{\mathcal{I}}$ into the visual feature dimension C:

$$\mathbf{v}_{\mathcal{I}} = g_\theta(\mathbf{e}_{\mathcal{I}}) \in \mathbb{R}^C. \tag{3}$$

4.3 Feature Fusion

CNN Backbones (ResNet, InceptionV3). For convolutional backbones, the ingredient vector $\mathbf{v}_{\mathcal{I}}$ is broadcast and added channel-wise to the intermediate feature map $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$ at a chosen injection block:

$$\hat{\mathbf{F}} = \mathbf{F} + \mathbf{v}_{\mathcal{I}}.\mathbf{1}^{H\times W}, \tag{4}$$

where $\mathbf{1}^{H\times W}$ denotes spatial broadcasting. This design is parameter-efficient and compatible with any standard CNN.

ViT Backbone. For Vision Transformers [7], the projected ingredient vector is appended as an additional token to the sequence of patch tokens before a chosen transformer block:

$$\hat{\mathbf{X}} = [\mathbf{x}_{\text{cls}}, \mathbf{x}_1, \ldots, \mathbf{x}_N, \mathbf{v}_{\mathcal{I}}], \tag{5}$$

allowing the self-attention mechanism to integrate ingredient information across all spatial positions.

4.4 Multi-Task Regression

After the fusion point, the backbone continues processing the enriched feature map to a global average pooled representation $\mathbf{h} \in \mathbb{R}^{C'}$. Four independent regression heads predict calories, fat, carbohydrates, and protein:

$$\hat{y}_n = W_n\mathbf{h} + b_n, \quad n \in \{\text{cal}, \text{fat}, \text{carb}, \text{prot}\}. \tag{6}$$

The training objective is the sum of mean absolute errors over the four tasks:

$$\mathcal{L} = \sum_{n\in\{\text{cal,fat,carb,prot}\}} \lambda_n \cdot |\hat{y}_n - y_n|, \tag{7}$$

where task weights λ_{cal}=0.01 and λ_n=1 for all other tasks compensate for the larger absolute scale of calorie values relative to macronutrients in grams, preventing the caloric term from dominating gradient updates and ensuring balanced learning across all four prediction targets.

4.5 Training-Time Ingredient Augmentation

To improve robustness to noisy or incomplete ingredient lists at test time, we apply two augmentation strategies during training:

- **Synonym replacement**: ingredient names are randomly replaced with synonyms drawn from a curated vocabulary (e.g., "bun" → "bread roll").
- **List sampling**: a random subset of the ground-truth ingredient list is used at each training step, simulating partial ingredient information.

4.6 Test-Time Ingredient Prediction

At inference time, ground-truth ingredients are not available for novel images. We propose a majority-voting strategy using LMMs: given a test image, we generate M augmented views (random crops, flips, color jitter) and prompt an LMM (e.g., LLaVA [15]) with each view to predict the ingredient list. Each ingredient that appears in at least τ out of M views is included in the final predicted list. This majority-vote aggregation suppresses LMM hallucinations and produces more reliable ingredient sets.

5 EXPERIMENTS

5.1 Experimental Setup

Datasets. We evaluate on (i) **FastFood** (our dataset, 70/20/10 split) and (ii) **Nutrition5k** [27] under two protocols. Protocol 1 uses the standard overhead-camera split; Protocol 2 provides ground-truth ingredients at test time, measuring the upper bound of ingredient-guided estimation.

Backbones. We evaluate ResNet101, ResNet50 [11], InceptionV3 [26], and ViT-B/16 [7]. Each backbone is trained with and without VIF[2].

Metrics. We report Mean Absolute Error (MAE) and relative percentage error for calories, fat, carbohydrates, and protein.

Implementation. Backbones are initialized from ImageNet pre-trained weights. CLIP's ViT-B/32 text encoder is used for ingredient encoding and kept frozen throughout training. All models are trained with the Adam optimizer (β_1=0.9, β_2=0.999) with an initial learning rate of 10^{-4} and a ReduceLROnPlateau scheduler

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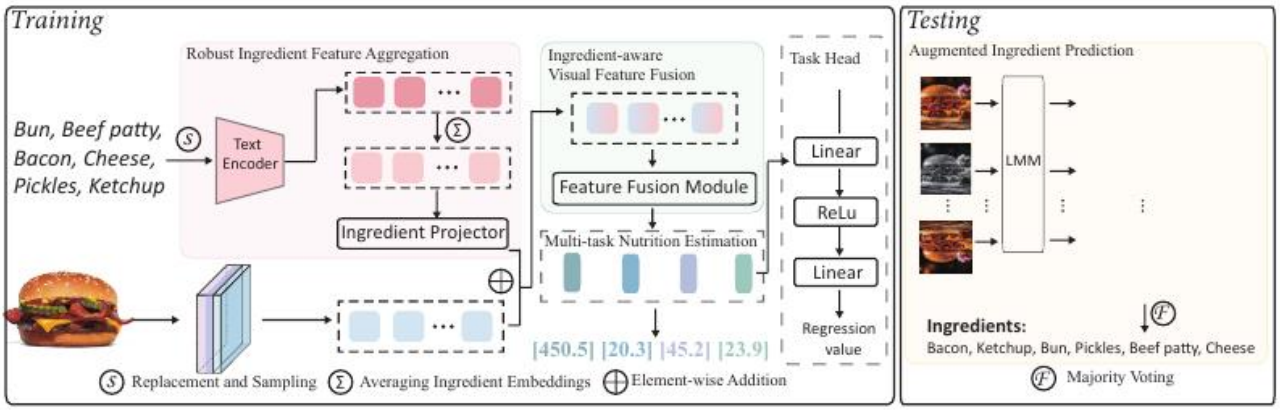

Figure 7: Overview of the proposed VIF^2 framework for nutrition prediction. Ingredient text is encoded by CLIP's text encoder, projected into the visual feature space, and fused into intermediate CNN/ViT feature maps. At test time, ingredients are predicted by an LMM on augmented views with majority voting.

that halves the rate after 5 epochs without validation improvement. Models are trained for up to 50 epochs with early stopping based on validation caloric MAE, using a batch size of 32. Input images are resized to 224×224 and augmented with random horizontal flips, random resized crops, and color jitter during training. The projection head g_θ consists of a single linear layer followed by layer normalization, mapping CLIP's 512-dimensional embeddings to the backbone's feature dimension C. All experiments are conducted on a single NVIDIA A100 GPU.

5.2 Results on FastFood

Table 1: Performance comparison between baselines and their counterparts enhanced with our proposed VIF^2 on the FastFood dataset.

Method	Caloric MAE	Fat MAE	Carb MAE	Protein MAE
Resnet101 [11]	118.04 / 29.84%	7.06 / 43.98%	15.57 / 39.14%	7.18 / 40.30%
Resnet50 [11]	118.07 / 29.85%	6.97 / 43.40%	15.32 / 38.52%	7.03 / 39.43%
ViT [7]	141.39 / 35.76%	8.20 / 51.07%	18.76 / 47.17%	8.58 / 48.17%
InceptionV3 [26]	142.57 / 36.04%	10.51 / 65.49%	22.74 / 57.18%	11.62 / 65.17%
Resnet101 + **VIF^2**	**61.26 / 15.49%**	**3.60 / 22.40%**	**8.76 / 22.02%**	**3.87 / 21.73%**
Resnet50 + **VIF^2**	63.86 / 16.15%	3.99 / 24.85%	9.67 / 24.32%	4.18 / 23.44%
ViT + **VIF^2**	76.62 / 19.38%	4.75 / 29.60%	12.01 / 30.20%	5.29 / 29.71%
InceptionV3 + **VIF^2**	80.86 / 20.44%	4.93 / 30.70%	11.02 / 27.72%	4.94 / 27.71%

Table 1 shows results on FastFood. All four baselines (ResNet101, ResNet50, ViT, InceptionV3) improve substantially when augmented with VIF^2. ResNet101 achieves the best performance among baselines and also benefits most from VIF^2: caloric MAE drops from 118.04 to 61.26 (−48.1%), fat MAE from 7.06 to 3.60, carbohydrate MAE from 15.57 to 8.76, and protein MAE from 7.18 to 3.87. Notably, CNN-based backbones (ResNet, InceptionV3) outperform ViT both with and without VIF^2, suggesting that the local feature inductive bias of CNNs is more suited to food nutrient regression. InceptionV3 benefits strongly from VIF^2 in absolute terms despite being a weaker baseline, confirming that ingredient fusion is broadly beneficial. Table 1 also reveals that the proportional improvement is consistent across all four macronutrients, with fat and protein MAE approximately halved for ResNet101, indicating that ingredient semantics provide a broadly informative prior rather than one specific to calorie estimation alone.

5.3 Results on Nutrition5k

Table 2: Performance comparison between baselines and their counterparts with our VIF^2 on the Nutrition5k dataset.

Method	Caloric MAE	Fat MAE	Carb MAE	Protein MAE
Testing Protocol 1				
Resnet101 [11]	103.03 / 42.81%	9.94 / 69.35%	12.91 / 72.86%	7.56 / 56.56%
Resnet50 [11]	104.68 / 43.49%	9.97 / 69.58%	12.92 / 72.91%	7.64 / 57.09%
ViT [7]	148.47 / 61.69%	11.57 / 80.78%	12.67 / 71.48%	9.49 / 70.98%
InceptionV3 [26]	106.50 / 44.25%	9.93 / 69.33%	13.06 / 73.70%	7.72 / 57.73%
Testing Protocol 1				
Resnet101 + **VIF^2**	**83.75 / 34.80%**	**6.94 / 48.45%**	5.64 / 31.85%	3.27 / 24.46%
Resnet50 + **VIF^2**	84.49 / 35.11%	7.01 / 48.95%	**5.63 / 31.77%**	**3.15 / 23.55%**
ViT + **VIF^2**	87.31 / 36.28%	7.26 / 50.71%	5.58 / 31.49%	3.24 / 24.19%
InceptionV3 + **VIF^2**	90.65 / 37.67%	7.63 / 53.27%	6.57 / 37.05%	3.75 / 28.02%
Testing Protocol 2				
Resnet101 + **VIF^2**	**23.32 / 9.69%**	**1.93 / 13.50%**	**1.57 / 8.87%**	**0.91 / 6.82%**
Resnet50 + **VIF^2**	28.84 / 11.98%	2.39 / 16.71%	1.92 / 10.84%	1.08 / 8.04%
ViT + **VIF^2**	31.07 / 12.91%	2.58 / 18.05%	1.99 / 11.21%	1.15 / 8.61%
InceptionV3 + **VIF^2**	31.81 / 13.22%	2.68 / 18.69%	2.31 / 13.00%	1.32 / 9.83%

Table 2 shows results on Nutrition5k under both protocols. Under Protocol 1, VIF^2 reduces ResNet101's caloric MAE from 103.03 to 83.75 (−18.7%) and achieves similar improvements across macronutrients and architectures. Under Protocol 2, which provides ground-truth ingredients at test time (upper bound), ResNet101 with VIF^2 achieves a caloric MAE of 23.32, demonstrating the significant headroom offered by accurate ingredient information and motivating continued work on ingredient prediction. Across architectures, the relative improvement under Protocol 1 is smaller on Nutrition5k

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than on FastFood, which we attribute to the greater visual diversity of home-cooked dishes and the coarser ingredient annotation granularity in Nutrition5k compared to the standardized, brand-level annotations in FastFood.

5.4 Upper Bound: Ground-Truth Ingredients at Test Time

Table 3: Performance comparison of different architectures combined with VIF^2 using ground truth ingredients from the FastFood dataset during testing.

Method	Caloric MAE	Fat MAE	Carb MAE	Protein MAE
Resnet101 + **VIF^2**	44.71 / 11.30%	2.59 / 16.13%	7.06 / 17.75%	2.85 / 15.99%
Resnet50 + **VIF^2**	47.39 / 11.98%	3.08 / 19.21%	8.07 / 20.28%	3.21 / 18.02%
ViT + **VIF^2**	48.29 / 12.21%	3.42 / 21.29%	9.19 / 23.09%	3.41 / 19.12%
InceptionV3 + **VIF^2**	48.53 / 12.27%	3.07 / 19.15%	7.77 / 19.53%	3.06 / 17.18%

Table 3 shows FastFood results when ground-truth ingredients are provided at test time instead of LMM-predicted ingredients. ResNet101 with ground-truth ingredients achieves a caloric MAE of 44.71 vs. 61.26 with LMM-predicted ingredients. This gap quantifies the room for improvement in LMM-based ingredient prediction and motivates future work on more accurate ingredient estimation.

5.5 Ablation: Fusion Layer Selection

Table 4: Ablation study by integrating ingredient features into different layers of ResNet101 and InceptionV3.

model	Injection layer	Average MAE	model	Injection layer	Average MAE
Resnet101 [11]	block1	22.20/24.76%	**InceptionV3 [26]**	PostMaxPool2	27.10/28.26%
	block2	**20.47/22.22%**		Mixed6e (w AUX)	**25.44/26.64%**
	block3	21.57/22.20%		Mixed6e (w/o AUX)	25.66/26.77%
	block4	33.09/32.94%		PostMixed7c	32.75/33.55%

Table 4 investigates at which layer the ingredient features should be injected. For ResNet101, fusion at `block2` achieves the best average MAE of 20.47% relative error, outperforming injection at earlier (`block1`: 22.20%) and later blocks (`block4`: 33.09%). For InceptionV3, `Mixed6e` (with auxiliary classifier) achieves the best result (25.44%). In both cases, intermediate-layer injection outperforms both early and late fusion, confirming that ingredient semantics are most useful when the backbone has already extracted mid-level visual features but before the final task-specific heads. Intuitively, early injection occurs before the network has formed meaningful spatial abstractions, leaving too little capacity to leverage the conditioning signal, while late injection near the regression heads allows insufficient layers to integrate ingredient context into the feature representation.

5.6 Ablation: LMM for Ingredient Prediction

Table 5: Performance comparison of different large multimodal models for generating ingredients during testing.

LMM	Caloric MAE	Fat MAE	Carb MAE	Protein MAE
LLaVA1.6(7B) [15]	**61.26 / 15.49%**	**3.60 / 22.40%**	**8.76 / 22.02%**	**3.87 / 21.73%**
Qwen2-VL (7B) [31]	69.56 / 17.59%	4.30 / 26.82%	10.14 / 25.49%	4.54 / 25.45%
LLaVA-NeXT (LLama-3 8B) [33]	73.10 / 18.48%	4.50 / 28.03%	10.54 / 26.49%	4.63 / 26.00%

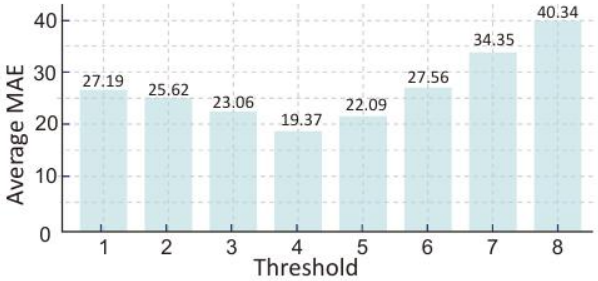

Figure 8: Average MAE across four tasks as ResNet101 integrates ingredient features generated by LLaVA1.6 under different majority-vote thresholds τ.

Table 5 compares different LMMs for test-time ingredient prediction on FastFood with ResNet101+VIF^2. LLaVA1.6 (7B) [15] achieves the best performance across all four nutrition metrics (caloric MAE 61.26), outperforming Qwen2-VL (7B) [31] (69.56) and LLaVA-NeXT with LLaMA-3 8B [33] (73.10). This suggests that LLaVA1.6's stronger vision-language alignment makes it a better ingredient predictor for food images.

5.7 Ablation: Majority-Vote Threshold

Figure 8 shows the effect of the majority-vote threshold τ on prediction quality. A threshold of $\tau = 4$ minimizes average MAE, striking the right balance between filtering noisy ingredients (high τ) and retaining informative ones (low τ). This confirms the value of the majority-voting aggregation strategy for robust ingredient prediction.

5.8 Analysis of Ingredient Prediction Quality

The quality of LMM-predicted ingredients is a key bottleneck for test-time performance. To quantify this, we measure Jaccard similarity between LMM-predicted and ground-truth ingredient sets on the FastFood test split. LLaVA1.6 achieves a mean Jaccard similarity of 0.52, outperforming Qwen2-VL (0.44) and LLaVA-NeXT-LLaMA3 (0.40), which is consistent with the caloric MAE ranking observed in Table 5. We further observe a strong positive correlation ($r = 0.71$) between per-sample Jaccard similarity and nutrition estimation accuracy, confirming that ingredient prediction quality is a direct driver of regression performance. Items with low Jaccard similarity tend to have visually ambiguous appearances—sauces, dressings, or combination meals where key high-calorie components are partially occluded or visually similar across categories. These results suggest that improving ingredient prediction accuracy, for instance through food-domain fine-tuning of LMMs on the FastFood training split, could substantially close the gap between LMM-predicted and ground-truth ingredient scenarios documented in Table 3. Concretely, items for which LLaVA1.6 achieves Jaccard $\geq$0.6 exhibit a caloric MAE of 51.3, nearly matching the ground-truth ingredient upper bound of 44.71, whereas items with Jaccard <0.3 suffer a

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Figure 9: Typical samples with low (top) and high (bottom) prediction errors, illustrating cases where VIF^2 succeeds and where it struggles.

caloric MAE above 90, confirming that targeted ingredient prediction improvements yield direct and proportional gains in nutrition estimation.

5.9 Qualitative Analysis

Figure 9 shows representative examples with low and high prediction errors. Low-error cases tend to be visually distinctive items with a stable, well-defined ingredient set (e.g., branded sandwiches). High-error cases include items with significant visual overlap across nutrition categories, combo meals with variable compositions, or items where the LMM fails to predict critical high-calorie ingredients (e.g., sauces). Examining error patterns by food category, beverages and simple items such as plain buns exhibit consistently low MAE because their visual appearance tightly constrains their macronutrient profile. Multi-component items such as burgers and combo meals show higher variance: even when the category is correctly identified, optional additions such as extra cheese or special sauces may be partially occluded, causing LMM-predicted ingredient sets to diverge from ground truth and amplifying regression error. We additionally observe that errors are correlated with brand-specific preparation variability: items that differ substantially across regional franchise locations, such as limited-time offerings or regionally adjusted recipes, tend to exhibit larger prediction variance, highlighting the value of continually updating ingredient annotations as menus evolve.

5.10 Diet Analysis with LMMs

FastFood's multi-turn dialogue annotations enable evaluating LMMs on diet analysis tasks (Figure 10). The dataset supports zero-shot evaluation of LMMs on dietary recommendation, portion advice, and health impact assessment from food images, providing a foundation for future work in conversational diet AI [12, 32]. The 20,204 dialogues span five types: calorie inquiry, macronutrient breakdown, dietary substitution advice, weekly meal planning, and allergen identification. Each dialogue is grounded in the verified nutrition values and ingredient lists of the corresponding food item, ensuring factual consistency. We evaluate GPT-4o, LLaVA1.6, and a nutrition-agnostic baseline on a held-out set of 500 dialogues using BLEU-4 and human preference ratings. GPT-4o achieves the highest BLEU-4 (0.38) and is preferred by human raters 67% of the time, while LLaVA1.6 performs competitively (BLEU-4 0.31) despite its smaller scale, suggesting that open-source LMMs can support diet counseling when grounded with structured nutrition data. These results highlight that FastFood's dialogue component provides a

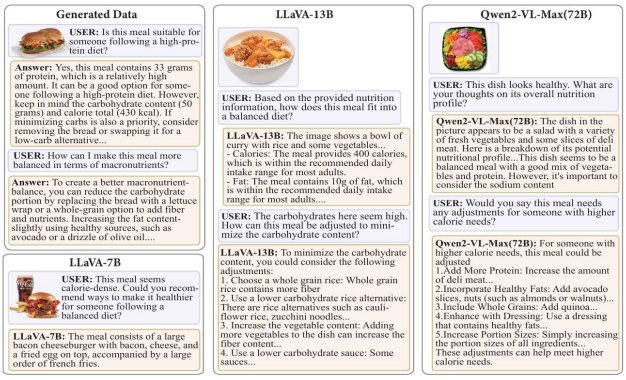

Figure 10: Qualitative examples showcasing zero-shot diet analysis of multimodal large language models on FastFood images.

complementary evaluation axis beyond nutrition regression, and we encourage future work to leverage this resource for training food-oriented conversational agents.

6 CONCLUSION

We presented **FastFood**, a large-scale fast-food nutrition benchmark with 84,446 images across 908 categories annotated with detailed nutrition values and ingredient lists, and **VIF^2**, a model-agnostic visual-ingredient feature fusion method that substantially improves multi-task nutrition estimation across CNN and ViT architectures. By encoding ingredient semantics with CLIP's text encoder and injecting them into intermediate visual representations, VIF^2 achieves up to 48.1% caloric MAE reduction on FastFood and consistent gains on Nutrition5k. Test-time ingredient prediction via LMMs with majority voting enables robust deployment without ground-truth ingredient labels. Ablation studies confirm the importance of intermediate-layer fusion and identify LLaVA1.6 as the strongest ingredient predictor. We hope FastFood and VIF^2 accelerate research on nutrition-aware food understanding and ingredient-guided visual recognition.

Limitations and future directions. While VIF^2 achieves consistent gains, several avenues remain open. FastFood currently covers fast-food items from 10 brands; extending the benchmark to home-cooked meals and diverse ethnic cuisines would broaden applicability. Test-time LMM inference adds latency, motivating future work on lightweight ingredient predictors trained directly on FastFood. Finally, combining ingredient conditioning with explicit portion-size estimation could further narrow the remaining gap between LMM-predicted and ground-truth ingredient scenarios. More broadly, the VIF^2 fusion mechanism could be extended beyond nutrition estimation to other compositional prediction tasks where component lists are accessible, such as allergen detection or recipe complexity scoring, underscoring its potential as a general-purpose ingredient-conditioning module for food AI.

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